

BS-CH101 / BS-CH201 — Chemistry-I

Semester I • 3L:1T:0P • 4 Credits

Module 7 Exam Notes: Organic Reactions and Drug Molecule Synthesis

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Organic Reactions and Drug Molecule Synthesis

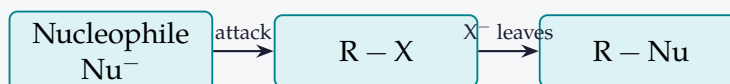
Organic reaction mechanism explains the step-by-step breaking and making of bonds during conversion of reactants into products. It identifies intermediates, reagents, and electron movement.

1.1 Substitution Reactions

In substitution, one atom or group is replaced by another. Nucleophilic substitution is common in alkyl halides.

Type	Main feature	Favoured by
S _N 1	Two-step, carbocation intermediate	Tertiary substrate, polar protic solvent
S _N 2	One-step backside attack	Methyl/primary substrate, strong nucleophile

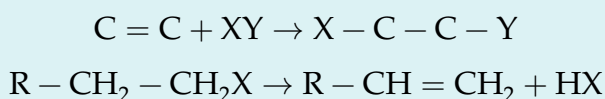
SN2 Backside Attack Concept



1.2 Addition and Elimination Reactions

Addition reactions add atoms across multiple bonds. Elimination reactions remove atoms from adjacent carbons to form a double bond.

Typical Patterns



1.3 Oxidation, Reduction, Cyclization and Ring Opening

Oxidation increases C-O, C-N, or C-X bonds or decreases C-H bonds. Reduction increases C-H bonds or removes electronegative atoms. Cyclization forms a ring; ring opening converts cyclic compounds into open-chain products.

Reaction	Meaning	Example idea
Oxidation	Gain of oxygen or loss of hydrogen	Alcohol to aldehyde/acid
Reduction	Gain of hydrogen or loss of oxygen	Carbonyl to alcohol
Cyclization	Ring formation	Intramolecular attack
Ring opening	Cleavage of a ring bond	Epoxide hydrolysis

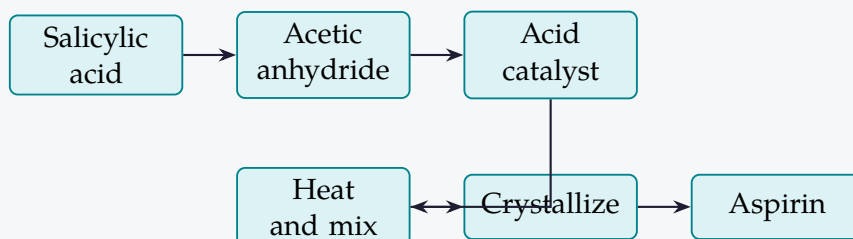
1.4 Synthesis of a Common Drug Molecule: Aspirin

Aspirin (acetylsalicylic acid) is prepared by acetylation of salicylic acid using acetic anhydride in acidic medium.

Aspirin Synthesis Reaction



Flow Diagram for Aspirin Preparation



1.5 Synthesis of a Common Drug Molecule: Paracetamol

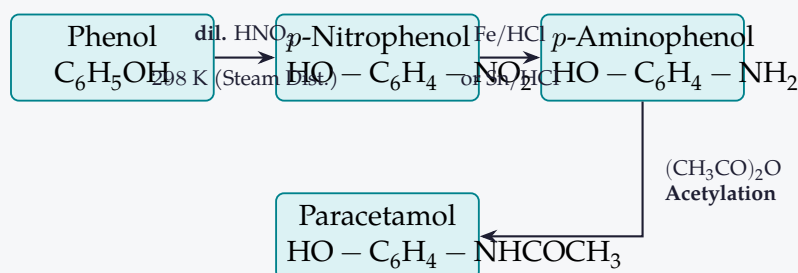
Paracetamol (acetaminophen or *p*-acetamidophenol) is a widely used analgesic (pain reliever) and antipyretic (fever reducer). It is synthesized from phenol via a three-step pathway.

Paracetamol Synthesis Reactions

The three steps involved in the industrial/laboratory synthesis are:

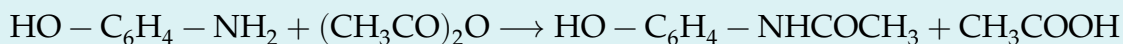
- Nitration of Phenol:** Phenol reacts with dilute nitric acid (HNO_3) at low temperature ($\sim 298 \text{ K}$) to yield a mixture of *o*-nitrophenol and *p*-nitrophenol. The para isomer is separated by steam distillation (since the ortho isomer is steam-volatile due to intramolecular hydrogen bonding).
- Reduction to *p*-Aminophenol:** *p*-nitrophenol is reduced to *p*-aminophenol using reducing agents like iron and hydrochloric acid (Fe/HCl) or tin and hydrochloric acid (Sn/HCl).
- Acetylation:** *p*-aminophenol is reacted with acetic anhydride ($(\text{CH}_3\text{CO})_2\text{O}$) in an aqueous or weakly acidic/basic medium. The amino group ($-\text{NH}_2$) undergoes selective acetylation to yield Paracetamol.

Reaction Scheme: Phenol to Paracetamol



Selective Acetylation Mechanism (Chemoselectivity)

In *p*-aminophenol, both the amino group ($-\text{NH}_2$) and the hydroxyl group ($-\text{OH}$) are nucleophilic. However, nitrogen is less electronegative than oxygen, making the lone pair on nitrogen much more nucleophilic and available for attack. Therefore, acetylation occurs exclusively at the nitrogen atom:

**Mechanism in Words:**

1. The nucleophilic nitrogen lone pair of *p*-aminophenol attacks one of the carbonyl carbons of acetic anhydride.
2. A tetrahedral intermediate is formed.
3. The carbonyl double bond is restored, and an acetate ion departs as the leaving group.
4. Deprotonation of the nitrogen atom gives Paracetamol (*p*-acetamidophenol) and acetic acid.

1.6 Mechanistic Exam Language

For mechanism-based answers, identify electrophile, nucleophile, leaving group, intermediate, and product. Use curved-arrow reasoning in words even when detailed curved arrows are not drawn.

Exam Points

- Compare $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ using step, intermediate, rate law, and stereochemistry.
- For addition and elimination, write the general reaction pattern.
- For oxidation/reduction, state change in C-H and C-O bonds.
- For aspirin synthesis, mention salicylic acid, acetic anhydride, acid catalyst, and acetylation.

2. Detailed Organic Reaction Notes and Synthesis Planning**2.1 Reaction Intermediates**

Organic reactions often proceed through reactive intermediates. Stability of intermediates decides product distribution and rate.

Intermediate	Nature	Stability trend
Carbocation	Electron-deficient carbon with positive charge	$3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$
Carbanion	Electron-rich carbon with negative charge	Stabilized by electron-withdrawing groups
Free radical	Species with unpaired electron	$3^\circ > 2^\circ > 1^\circ > \text{CH}_3\cdot$
Carbene	Neutral divalent carbon species	Highly reactive; addition to alkenes

2.2 Detailed Comparison of $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$

Point	$\text{S}_{\text{N}}1$	$\text{S}_{\text{N}}2$
Mechanism	Two-step	One-step concerted
Rate law	$k[\text{RX}]$	$k[\text{RX}][\text{Nu}^-]$
Intermediate	Carbocation	No intermediate
Substrate	$3^\circ > 2^\circ > 1^\circ$	methyl $> 1^\circ > 2^\circ$
Solvent	Polar protic	Polar aprotic
Stereochemistry	Racemization possible	Inversion of configuration

Solved Concept: Choosing $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$

Tertiary butyl chloride reacts with water mainly by $\text{S}_{\text{N}}1$ because a stable tertiary carbocation can form and water is a weak nucleophile in polar protic medium. Methyl bromide reacts with OH^- mainly by $\text{S}_{\text{N}}2$ because backside attack is unhindered.

2.3 Electrophilic Addition to Alkenes

Alkenes undergo electrophilic addition because the pi bond is electron rich. Addition of HX follows Markovnikov rule in normal ionic addition.

Markovnikov rule: In addition of HX to an unsymmetrical alkene, hydrogen attaches to the carbon already having more hydrogen atoms, and X attaches to the more substituted carbon.

Addition of HBr to Propene



The major product is 2-bromopropane due to formation of the more stable secondary carbocation.

2.4 Elimination: $\text{E}1$ and $\text{E}2$

Elimination reactions form alkenes. $\text{E}1$ proceeds through carbocation; $\text{E}2$ occurs in one step with base removing beta-hydrogen as leaving group departs.

Point	E1	E2
Steps	Two-step	One-step
Intermediate	Carbocation	None
Base strength	Weak base possible	Strong base required
Substrate	Tertiary favoured	Primary, secondary, tertiary possible
Stereochemical need	No strict anti requirement	Anti-periplanar H and leaving group preferred

2.5 Oxidation and Reduction Reagents

Remembering reagent function is more useful than memorizing long mechanisms for basic engineering chemistry.

Reagent	Type	Common use
KMnO ₄	Strong oxidizing agent	Alkene oxidation, alcohol oxidation
K ₂ Cr ₂ O ₇ /H ⁺	Oxidizing agent	Primary alcohol to acid, secondary alcohol to ketone
LiAlH ₄	Strong reducing agent	Carbonyl and acid derivatives to alcohols
NaBH ₄	Mild reducing agent	Aldehyde/ketone to alcohol
H ₂ /Ni	Catalytic reduction	Alkene to alkane

2.6 Cyclization and Ring Opening

Cyclization is favoured when the reacting ends of a molecule can form a stable five- or six-membered ring. Ring opening is important for strained rings like epoxides.

Cyclization and Ring Opening Concept



2.7 Aspirin Synthesis: Mechanism and Purification

In aspirin preparation, salicylic acid provides the phenolic –OH group and acetic anhydride provides the acetyl group. Acid catalyst increases electrophilicity of acetic anhydride. The product is purified by recrystallization.

1. Protonation of acetic anhydride increases electrophilic character of carbonyl carbon.
2. Phenolic oxygen of salicylic acid attacks the carbonyl carbon.
3. Acetate group leaves and deprotonation gives aspirin.
4. Cold water helps crystallize aspirin because aspirin is less soluble in cold water.

Aspirin Yield Calculation

If theoretical yield is 5.00 g and actual yield is 3.80 g:

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

$$\% \text{ yield} = \frac{3.80}{5.00} \times 100 = 76\%$$

Organic Reaction Answer Checklist

1. Identify substrate, reagent, nucleophile/electrophile, and leaving group.
2. Classify the reaction as substitution, addition, elimination, oxidation, reduction, cyclization, or ring opening.
3. Write rate law or regioselectivity rule where applicable.
4. Mention major product and reason for its formation.
5. For drug synthesis, include reactants, catalyst, reaction type, purification, and yield idea.

2.8 Reaction Map for Basic Organic Transformations

Most exam questions ask how one functional group changes into another. The following map gives the common first-year transformations.

Conversion	Typical reagent	Reaction class
Alkene to alkane	H ₂ /Ni	Reduction / hydrogenation
Alkene to alcohol	H ₂ O/H ⁺	Addition
Alcohol to alkene	conc. H ₂ SO ₄ , heat	Elimination / dehydration
Primary alcohol to acid	KMnO ₄ or K ₂ Cr ₂ O ₇ /H ⁺	Oxidation
Aldehyde to alcohol	NaBH ₄	Reduction
Alkyl halide to alcohol	aqueous KOH	Nucleophilic substitution

2.9 Curly-Arrow Logic in Words

Even when curved arrows are not drawn, the answer should describe electron flow clearly:

1. A nucleophile donates an electron pair.
2. An electrophile accepts an electron pair.
3. A leaving group departs with the bonding electron pair.
4. Formation of a stable intermediate or transition state controls the major product.

Worked Mechanism in Words: Alcohol to Alkene

In acid-catalysed dehydration of an alcohol, the —OH group is first protonated to form a better leaving group, water. Water leaves to form a carbocation. A base removes a beta-hydrogen, and a double bond forms. The major alkene is usually the more substituted alkene according to Zaitsev rule.

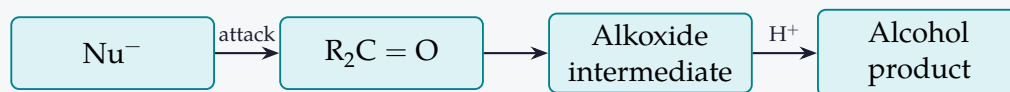
2.10 Zaitsev and Hofmann Elimination

When more than one alkene can form, regioselectivity matters.

Rule	Major product	Typical condition
Zaitsev rule	More substituted alkene	Small strong base or normal elimination
Hofmann rule	Less substituted alkene	Bulky base or quaternary ammonium elimination

2.11 Addition to Carbonyl Compounds

The carbonyl group is polar. Carbon is electrophilic and oxygen is nucleophilic/basic. Nucleophiles attack the carbonyl carbon to form addition products.

Carbonyl Addition Concept**2.12 Green Chemistry Note for Drug Synthesis**

Engineering chemistry answers may include practical synthesis quality. A good drug synthesis should be high-yielding, selective, safe, and easy to purify.

Green chemistry point	Meaning in aspirin synthesis
Atom economy	Prefer maximum incorporation of reactant atoms into product
Safer solvent	Use minimum harmful solvent during purification
Energy efficiency	Avoid unnecessary heating time
Waste reduction	Reduce excess reagent and by-product handling
Product purity	Recrystallization improves purity for medicinal use

2.13 Tests for Functional Groups

Short practical-style questions may ask how a functional group is detected.

Functional group	Test	Positive observation
Alkene	Bromine water	Decolorization
Aldehyde	Tollens' reagent	Silver mirror
Carboxylic acid	Sodium bicarbonate	Brisk effervescence of CO ₂
Phenol	Ferric chloride test	Violet or coloured complex
Alcohol	Esterification	Fruity smell of ester

Previous-Year Style Questions

1. Compare S_N1 and S_N2 mechanisms with stereochemical outcome.
2. Explain electrophilic addition to propene using Markovnikov rule.
3. Compare E1 and E2 elimination reactions.
4. Write short notes on oxidation and reduction reagents in organic chemistry.
5. Describe aspirin synthesis, purification and percentage yield calculation.

Quick Revision — Module 7

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Key Formulas at a Glance

Formula / Rule	Meaning
$\text{rate}_{S_N1} = k[\text{RX}]$	First-order nucleophilic substitution.
$\text{rate}_{S_N2} = k[\text{RX}][\text{Nu}^-]$	Second-order nucleophilic substitution.
Addition pattern	$\text{C} = \text{C} + \text{XY} \rightarrow \text{X} - \text{C} - \text{C} - \text{Y}$ General addition across a multiple bond.
Elimination pattern	$\text{R} - \text{CH}_2 - \text{CH}_2\text{X} \rightarrow \text{R} - \text{CH} = \text{CH}_2 + \text{HX}$ Removal of HX to form an alkene.
Aspirin synthesis	Salicylic acid + Acetic anhydride $\rightarrow$ Aspirin + Acetic acid Acetylation reaction for aspirin preparation.
Paracetamol synthesis	p-aminophenol + $(\text{CH}_3\text{CO})_2\text{O} \rightarrow$ Paracetamol + CH_3COOH Selective N-acetylation of p-aminophenol.

One-Line Definitions

- **Nucleophile:** Electron-rich species that attacks electron-poor centre.
- **Electrophile:** Electron-deficient species that accepts electron pair.
- **Leaving group:** Group that departs with electron pair.
- **Substitution:** Replacement of one group by another.
- **Addition:** Addition of atoms across a multiple bond.
- **Elimination:** Removal of groups to form a multiple bond.
- **Acetylation:** Introduction of acetyl group, $\text{CH}_3\text{CO}-$.

Frequently Asked / Viva Questions

Q1. What is an organic reaction mechanism?

It is the stepwise description of bond breaking and bond formation.

Q2. Define nucleophile.

A nucleophile is an electron-rich species that donates an electron pair.

Q3. Define electrophile.

An electrophile is an electron-deficient species that accepts an electron pair.

Frequently Asked / Viva Questions (contd.)

Q4. Compare S_N1 and S_N2 briefly.

S_N1 is two-step and forms carbocation; S_N2 is one-step backside attack.

Q5. What is an addition reaction?

A reaction in which atoms or groups add across a double or triple bond.

Q6. What is an elimination reaction?

A reaction in which small molecules are removed to form a multiple bond.

Q7. How is oxidation recognized in organic chemistry?

Increase in C-O/C-X bonds or decrease in C-H bonds.

Q8. How is reduction recognized in organic chemistry?

Increase in C-H bonds or decrease in C-O/C-X bonds.

Q9. What is cyclization?

Intramolecular reaction leading to ring formation.

Q10. Name reactants used for aspirin synthesis.

Salicylic acid and acetic anhydride in presence of acid catalyst.

Q11. Name the starting material and reagents used for Paracetamol synthesis.

Phenol is the starting material. Reagents are dilute HNO_3 (nitration), followed by steam distillation to isolate the *p*-nitrophenol isomer, then reduction using Fe/HCl or Sn/HCl, and finally acetylation using acetic anhydride.

Q12. In Paracetamol synthesis, why does acetylation occur at the $-NH_2$ group rather than the $-OH$ group of *p*-aminophenol?

Nitrogen is less electronegative than oxygen, making the lone pair on the amino group ($-NH_2$) more nucleophilic and reactive than the lone pair on the hydroxyl group ($-OH$). This allows selective N-acetylation under controlled conditions.

Summary Reference Table

Reaction type	Core change	Exam clue
Substitution	Group replacement	Leaving group present
Addition	Break pi bond, add groups	Alkene/alkyne reactant
Elimination	Remove small molecule	Alkene product
Oxidation/reduction	Change in C-O and C-H bonds	Reagent decides direction
Drug synthesis	Planned sequence	Reactants, catalyst, product